Milestone 1 — Problem Definition & Wing Modelling (Step 1)

Project: Physics-Informed AI-Based Digital Twin for Structural Health Monitoring of Composite UAV Wings **Workflow option:** B — in-house FSDT MATLAB solver → AI digital twin **Deliverable of this step:** *"Geometry & Inputs Ready"* — every modelling decision fixed, justified and coded in `config/wing_baseline_config.m` **Script that reproduces every number below:** `milestone1/run_milestone1.m` (log: `results/data/M1_summary.txt`)

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1. Purpose of Step 1

The digital twin can only be as good as the physical model under it. Step 1 answers six questions:

1. **What structure** are we modelling? (geometry, idealisation)
2. **What is it made of?** (lamina and core properties)
3. **How is it built?** (stacking sequence → laminate stiffness)
4. **How is it supported?** (boundary conditions)
5. **What loads act on it?** (flight envelope → design load cases)
6. **How will it be discretised?** (finite-element model)

Every answer is stored in **one struct `cfg`**. The DOE of Milestone 3 then generates thousands of variants just by perturbing fields of `cfg`.

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2. Coordinate system and idealisation

Axis	Direction	Origin
x	span-wise, root → tip	wing root (fuselage side)
y	chord-wise, leading edge (LE) → trailing edge (TE)	leading edge
z	up (lift direction)	laminate mid-surface

Idealisation: the semi-wing is a **cantilever sandwich plate**, clamped at the root (x = 0) and free on the other three edges.

Assumption	Justification	Upgrade path (later)
Constant-thickness plate instead of a curved airfoil shell	Bending stiffness is matched to the real airfoil skins (Section 4). This is the standard "equivalent plate" wing idealisation used in aeroelasticity and SHM literature.	Variable-thickness elements follow the airfoil

Assumption	Justification	Upgrade path (later)
Foam-core sandwich, no discrete spars	Typical construction of small composite UAV wings (foam core + carbon skins)	Add spar caps as stiffened element strips
Root perfectly clamped	Fuselage attachment is much stiffer than the wing	Root rotational springs (from tests)
Linear elastic, small deflection	Tip deflection < 1 % of span at limit load (Milestone 2)	Geometric non-linearity is not needed
Quasi-steady aerodynamics	Design gust is slow compared with the 1st mode ($t_g/T_1 = 6$)	Unsteady (Küssner/Wagner) aero for flutter

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3. Geometry

Parameter	Symbol	Value
Semi-span	b	1.000 m
Root chord	c_r	0.250 m
Taper ratio	λ = c_t/c_r	1.0 (rectangular)
Leading-edge sweep	Λ_LE	0°
Airfoil	—	NACA 0012 (t/c = 0.12, t_max = 30 mm)
Full wing area	S	0.500 m²
Aspect ratio	AR = (2b)²/S	8.0

The mesh generator (`src/mesh/wing_mesh.m`) already supports taper and sweep through the map

$$x = \xi\, b, \qquad c(x) = c_r\,[1 - (1-\lambda)\,\xi], \qquad y = x\tan\Lambda_{LE} + \eta\, c(x), \qquad (\xi,\eta) \in [0,1]^2 .$$

So geometry can become a DOE variable later without any code change.

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4. Equivalent sandwich depth (from the real airfoil)

Problem. A plate has a constant thickness, but the airfoil does not. Which core thickness gives the *same bending stiffness* as skins wrapped around a NACA 0012?

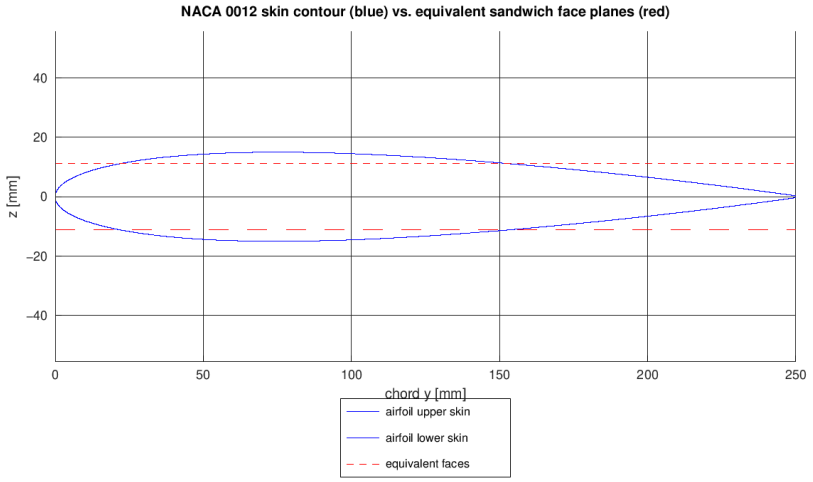
Theory. The NACA 4-digit half-thickness is

$$y_t(s) = 5\,\tau(0.2969\sqrt{s} - 0.1260\,s - 0.3516\,s^2 + 0.2843\,s^3 - 0.1015\,s^4)c, \qquad s = x/c,\ \tau = t/c.$$

- Skins of thickness t_f following the contour have, about the chord line, a second moment of area $I_{airfoil} = 2\,t_f \int_0^c y_t^2\, dx$.
- A flat sandwich whose face centroids are a distance d apart has $I_{sandwich} = 2\,t_f\, c\,(d/2)^2$.
- Equating the two gives the **bending-equivalent depth**

$$d_{eq} = 2\sqrt{\tfrac{1}{c}\int_0^c y_t^2\, dx}.$$

Result. For NACA 0012, c = 250 mm: **d_eq = 22.26 mm**. The mean airfoil thickness is 20.55 mm and the maximum is 30 mm. Chosen core: **t_c = 21.75 mm** + face 0.5 mm → **d = 22.25 mm (99.9 % of d_eq)**.



Equivalent sandwich

> **Why this matters.** If we had used the *maximum* thickness (30 mm), bending stiffness would be over-estimated by $(30/22.3)^2 \approx 1.8\times$. Frequencies would then be ~35 % too high, and every damage-sensitive feature the AI learns would be biased.

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5. Materials — orthotropic elasticity

5.1 Generalised Hooke's law for one ply

A unidirectional ply is **orthotropic**: it has three planes of material symmetry. In the material axes (1 = fibre, 2 = transverse, 3 = thickness), the compliance form is

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{13}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{Bmatrix}, \qquad \frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2} \text{ (Maxwell–Betti reciprocity).}$$

Plane-stress reduction ($\sigma_3 = 0$, valid for thin plies) gives the *reduced stiffness* used by all plate theories (`src/materials/ply_stiffness.m`):

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}}, \quad Q_{66} = G_{12}, \qquad Q_{44} = G_{23},\ Q_{55} = G_{13}.$$

FSDT keeps the two **transverse shear** stiffnesses Q_{44} and Q_{55}; classical theory throws them away. This is the whole reason for FSDT (Section 7).

5.2 Property database (`src/materials/material_library.m`)

Property	T300/5208 (skins)	Rohacell 51 WF (core)	AS4/3501-6 (DOE)	E-glass/epoxy (DOE)
E_1 [GPa]	181	0.075	147	38.6
E_2 [GPa]	10.3	0.075	10.3	8.27
G_{12} [GPa]	7.17	0.024	7.0	4.14
G_{13} [GPa]	7.17	0.024	7.0	4.14
G_{23} [GPa]	3.46	0.024	3.7	2.95
ν_{12}	0.28	0.30	0.27	0.26
ρ [kg/m³]	1600	52	1580	1800
X_t / X_c [MPa]	1500 / 1500	1.6 / 0.8	2280 / 1725	1062 / 610
Y_t / Y_c [MPa]	40 / 246	—	57 / 228	31 / 118
S_{12} [MPa]	68	0.8 (shear)	76	72

G_{23} of a transversely isotropic ply. The 2–3 plane is isotropic, so $G_{23} = E_2 / [2(1+\nu_{23})]$. With $\nu_{23} \approx 0.49$ for carbon/epoxy, $G_{23} = 10.3/2.98 =$ **3.46 GPa**. This value comes from the theory, so it replaces the rounded 3.0 GPa I quoted in the planning stage.

Foam core. For PMI foam, $E/(2G) - 1 = 0.56 > 0.5$, so it cannot be a perfectly isotropic solid. It is therefore stored as an orthotropic layer with independent E = 75 MPa and G = 24 MPa (Rohacell WF datasheet).

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6. Laminate — Classical Lamination Theory (CLT)

6.1 Ply rotation

A ply at angle θ to the x-axis (span) has transformed stiffness

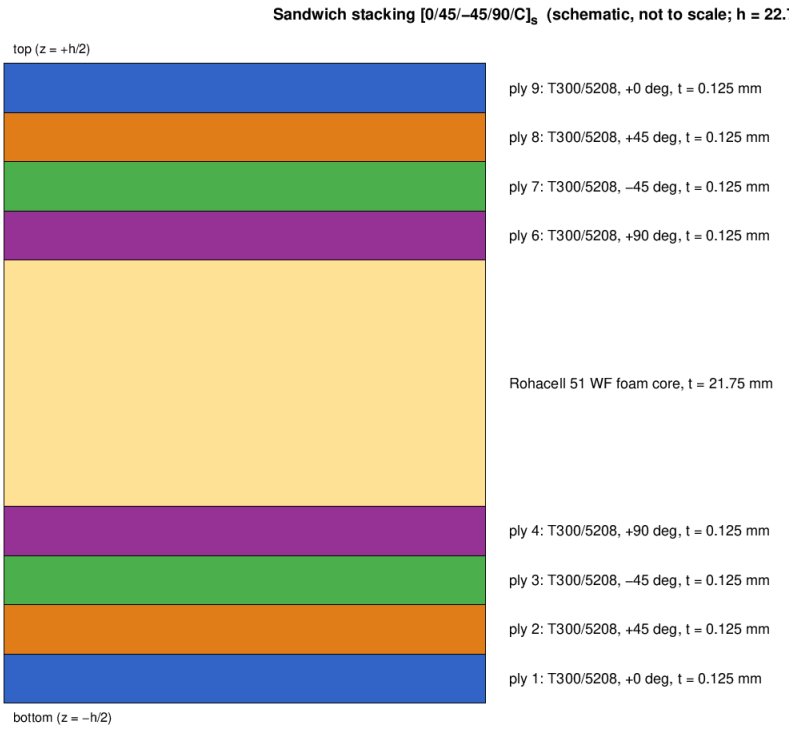
$$\bar{\mathbf{Q}} = \mathbf{T}_\varepsilon^T \, \mathbf{Q} \, \mathbf{T}_\varepsilon, \qquad \mathbf{T}_\varepsilon = \begin{bmatrix} c^2 & s^2 & cs \\ s^2 & c^2 & -cs \\ -2cs & 2cs & c^2 - s^2 \end{bmatrix}, \quad c = \cos\theta,\ s = \sin\theta,$$

using engineering shear strain. The form $T^T Q T$ follows from equal strain energy in both axis systems. The transverse shear stiffness rotates as a 2-D tensor:

$$\bar{\mathbf{Q}}_s = \mathbf{R}^T \mathbf{Q}_s \mathbf{R}, \qquad \mathbf{R} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}.$$

6.2 Stacking sequence (chosen)

[0/45/−45/90/CORE/90/−45/45/0] — quasi-isotropic carbon faces on a foam core, symmetric about the mid-plane.



Layup

Choice	Reason
0° plies outermost	They carry span-wise bending most efficiently (largest z^2)
±45° pair	Torsional stiffness and in-plane shear; balanced, so $A_{16} = A_{26} = 0$
90° next to the core	Chord-wise stiffness; protects the core interface
Symmetric	B = 0, so no bending–extension coupling and no warping on cure

6.3 Stiffness matrices

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\varepsilon}^0 \\ \boldsymbol{\kappa} \end{Bmatrix}, \qquad (\mathbf{A}, \mathbf{B}, \mathbf{D}) = \sum_{k=1}^{N} \bar{\mathbf{Q}}_k \left(z_k - z_{k-1}, \frac{z_k^2 - z_{k-1}^2}{2}, \frac{z_k^3 - z_{k-1}^3}{3}\right).$$

Computed values (`src/materials/laminate_section.m`):

Matrix	Value	Comment
A [MN/m]	$A_{11} = A_{22} = 78.16$, $A_{12} = 23.15$, $A_{66} = 27.40$, $A_{16} = 0$	Quasi-isotropic: $A_{66} = (A_{11} - A_{12})/2$ ✓
B [N]	$\max \lvert B_{ij} \rvert = 2 \cdot 10^{-11}$ (round-off)	Symmetric laminate ✓
D [N·m]	$D_{11} = 9703$, $D_{22} = 9345$, $D_{12} = 2819$, $D_{66} = 3348$, $D_{16} = D_{26} = 29.8$	$D_{11} > D_{22}$ because the 0° plies are outermost

Cross-check with sandwich theory (Allen, 1969): D ≈ E_f t_f d²/2 = 8623 N·m. CLT gives 9703 N·m. The 12 % difference is real: thin-face theory treats each face as one smeared layer, while CLT sees the 0° plies at the outer surface.

> **D_{16}, $D_{26} \neq 0$.** Even a “balanced” ±45 pair gives bending–twisting coupling, because the +45 and −45 plies sit at different z. The FE solver keeps the full D, so this coupling appears in the twist results of Milestone 2.

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7. Transverse shear and the shear-correction factor

7.1 Why the textbook k = 5/6 is wrong for this wing

FSDT assumes a **constant** shear strain through the thickness. The real shear stress follows from 3-D equilibrium, ∂τ_xz/∂z = −∂σ_x/∂x. Integrating layer by layer gives

$$\tau_{xz}(z) = -\frac{Q_x}{D_x} g(z), \qquad g(z) = \int_{-h/2}^{z} E(\zeta)\,(\zeta - z_n)\,d\zeta, \qquad E = \bar{Q}_{11}(z),$$

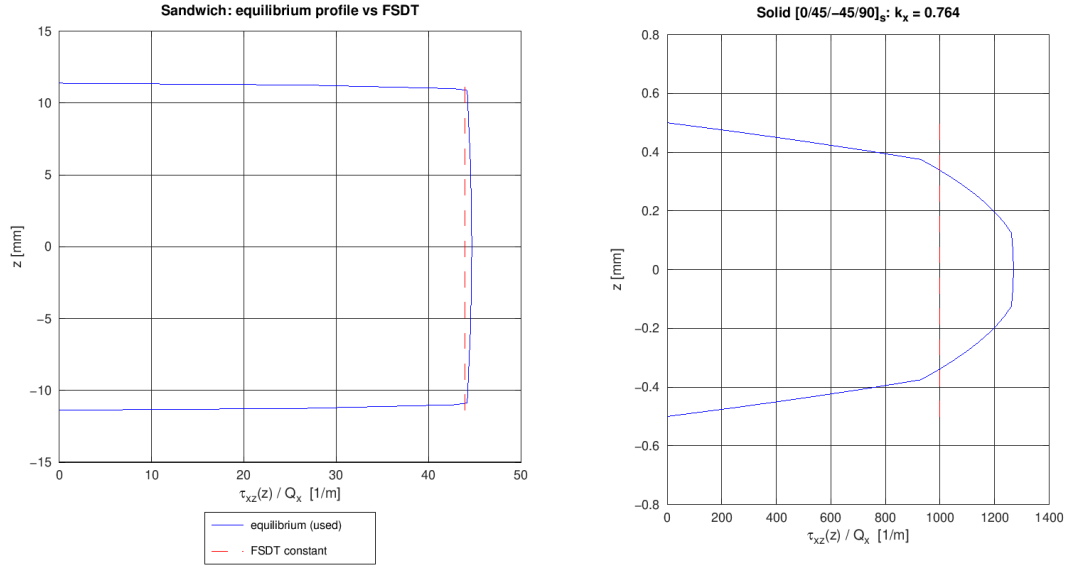
where z_n is the neutral axis. Equating the shear strain energy of this exact distribution with the FSDT energy Q_x²/(2k∫G dz) gives the **energy-equivalent shear correction factor** (Whitney 1973; Vlachoutsis 1992):

$$k_x = \frac{D_x^2}{\int_{-h/2}^{h/2} G(z)\,dz \int_{-h/2}^{h/2} \frac{g(z)^2}{G(z)}\,dz}, \qquad G = \bar{Q}_{55}.$$

`src/materials/shear_correction.m` evaluates this integral exactly. g² is quartic in each ply, so 3-point Gauss integration is exact.

Laminate	k_x	k_y
Homogeneous plate	**0.8333 = 5/6** (proves the formula)	0.8333
Solid $[0/45/-45/90]_s$, 8 plies	0.764	0.800
Our sandwich	**0.0954**	**0.0919**

7.2 Physical meaning



Shear profiles

In a sandwich the stiff faces carry almost no shear: the shear stress is nearly constant **in the soft core** and drops to zero across the faces. The correct shear stiffness is therefore

$$S = \frac{G_c\, d^2}{t_c} \quad \text{(Allen 1969)} = 0.546 \text{ MN/m},$$

Model	A_s,11 [MN/m]	Ratio to sandwich theory
FSDT, energy k (used)	0.556	**1.02** ✓
FSDT, k = 5/6	4.86	**8.9 ×**

> **Key insight for the project.** With k = 5/6 the model would make the core 9× stiffer in shear. A core debond or core crush is exactly a loss of core shear stiffness, so the AI would be trained on a structure that hides the very damage we want to detect. The energy-equivalent k is therefore a correctness requirement, not a refinement.

7.3 Mass properties

$$(I_0, I_1, I_2) = \sum_k \rho_k \left(z_k - z_{k-1},\ \frac{z_k^2 - z_{k-1}^2}{2},\ \frac{z_k^3 - z_{k-1}^3}{3}\right)$$

- I_0 = **2.731 kg/m²**.
- I_1 = 0 (symmetric).
- I_2 gives the rotary inertia, which FSDT keeps.

Semi-wing mass = **0.683 kg**. Both wings together are 22.8 % of MTOW, typical for a minimum-gauge small UAV.

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8. Flight envelope (V–n diagram) and design load factors

8.1 Aircraft data

m	W	W/S	C_Lmax / C_Lmin	V_C	V_D	n_+ / n_- manoeuvre
6 kg	58.9 N	117.7 N/m²	1.2 / −1.0	25 m/s	32 m/s (≥ 1.25 V_C)	+3.8 / −1.5

8.2 Theory

Stall boundary:

$$n_s(V) = \frac{\rho V^2 S C_{L\,\max}}{2W} \Rightarrow V_{S1} = \sqrt{\frac{2W}{\rho S C_{L\,\max}}} = 12.66 \text{ m/s}, \qquad V_A = V_{S1}\sqrt{n_+} = 24.67 \text{ m/s}.$$

Gust loads, Pratt–Walker formula (CS-23 style):

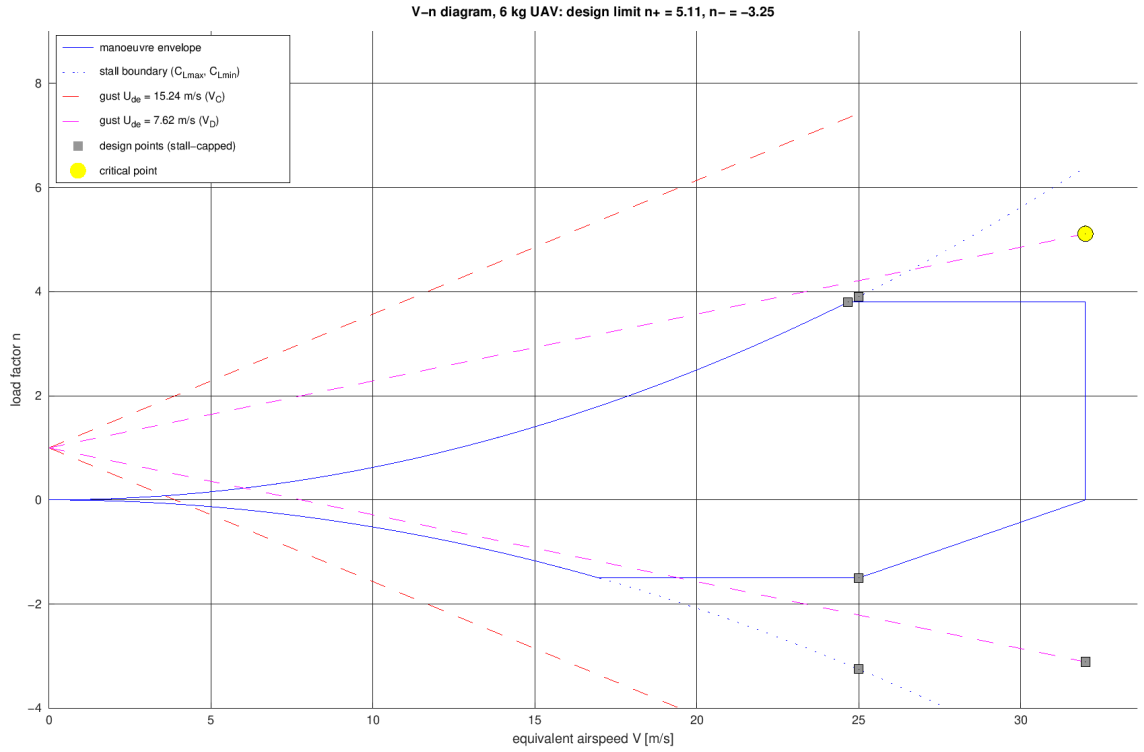
$$a = \frac{2\pi AR}{2 + \sqrt{AR^2 + 4}}, \qquad \mu = \frac{2W/S}{\rho \bar{c} a g}, \qquad K_g = \frac{0.88\mu}{5.3 + \mu}, \qquad \Delta n = \frac{\rho U_{de} V a K_g}{2W/S}.$$

a	μ	K_g	Δn at V_C (U = 15.24 m/s)	Δn at V_D (U = 7.62 m/s)
4.906 /rad	15.97	0.661	6.43	4.11

Stall cap: the wing cannot produce more lift than C_Lmax at a given speed, so each gust point is limited to n ≤ n_s(V).

8.3 Result

Design point	V [m/s]	n
Manoeuvre +	24.7	+3.80
Manoeuvre −	25.0	−1.50
Gust + at V_C (stall-capped from 7.43)	25.0	+3.90
Gust + at V_D	**32.0**	**+5.11 ← critical**
Gust − at V_C	25.0	−3.25
Gust − at V_D	32.0	−3.11



V-n diagram

> **Engineering finding.** Light UAVs have low wing loading, so they are **gust-critical, not manoeuvre-critical**: the design limit factor 5.11 exceeds the manoeuvre limit 3.8 by 34 %. A model built only on the manoeuvre limit would be un-conservative.

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9. Aerodynamic load model and load cases

9.1 Span-wise lift — Schrenk's method (NACA TM-948)

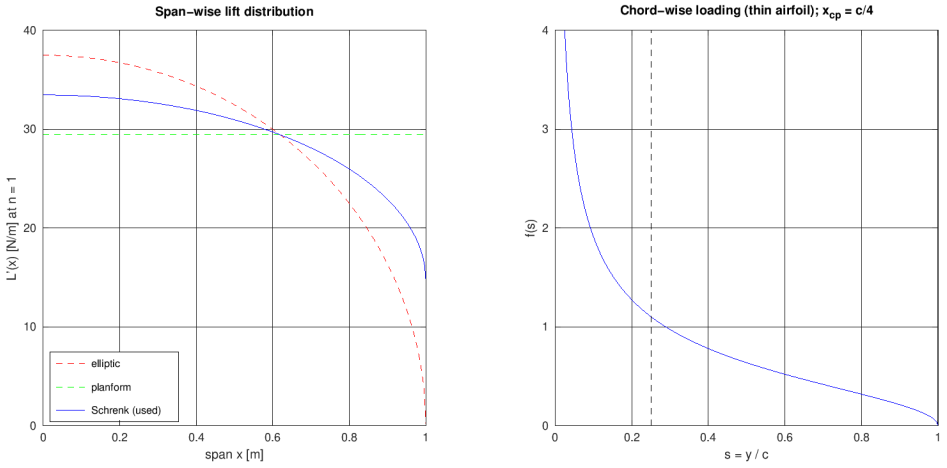
$$L'_e(x) = \frac{4L}{\pi b}\sqrt{1-(x/b)^2}, \qquad L'_p(x) = \frac{L\,c(x)}{S_{semi}}, \qquad L'(x) = \tfrac{1}{2}[L'_e(x) + L'_p(x)].$$

9.2 Chord-wise lift — thin-airfoil theory

For a thin airfoil at incidence, the pressure difference is

$$f(s) = \frac{2}{\pi}\frac{\sqrt{1-s}}{s}, \qquad \int_0^1 f\,ds = 1, \qquad \int_0^1 s\,f\,ds = \tfrac{1}{4}.$$

- The centre of pressure is exactly at the **quarter chord** (the aerodynamic centre). It lies ahead of the plate's elastic axis at mid-chord, so lift also **twists the wing nose-up**. This is the physical seed of divergence, and it gives the AI a torsional signal to learn from.
- Pressure field: p(x,y) = L′(x) f(s) / c(x).



Lift distribution

9.3 Numerical detail: the leading-edge singularity

f(s) ~ s^(−1/2) is integrable but singular at the LE, so ordinary Gauss quadrature converges slowly:

Gauss points	Total lift recovered	Centre-of-pressure arm (exact 0.0625 m)
6 × 6 (plain)	97.3 %	0.0608 m
24 × 24 (plain)	99.3 %	0.0621 m

Remedy (`src/loads/pressure_load.m`): in LE elements, use the substitution $\eta = -1 + 2t^2$, $d\eta = 4t\,dt$. The $1/\sqrt{s}$ singularity is cancelled by the Jacobian. With only 4 points the result is **100.003 % of the lift and an arm of 0.062496 m (exact to 10^{-5})**.

9.4 Inertia relief

At load factor n, the wing's own mass pushes down while lift pushes up:

$$\mathbf{F} = n\frac{W}{2}\hat{\mathbf{F}}_{lift} \;-\; n g \mathbf{M}\,\mathbf{r}_z,$$

where M is the consistent mass matrix and r_z the unit vertical rigid-body vector. For LC1 this reduces the root bending moment by about 25 % (13.6 → 10.25 N·m).

9.5 Design load cases (`src/loads/wing_load_cases.m`)

Case	n	Net F_z [N]	Root bending M_x [N·m]	Torque about mid-chord [N·m]
LC1 cruise 1 g	1.00	22.73	10.25	1.84
LC2 manoeuvre +	3.80	86.38	38.97	6.99
LC3 manoeuvre −	−1.50	−34.10	−15.38	−2.76
LC4 critical gust	**5.11**	**116.22**	**52.42**	**9.40**
LC5 ultimate (1.5 × LC4)	7.67	174.33	78.64	14.11
LC6 ground tip test (50 N, mid-chord)	—	50.00	50.00	0.00

LC6 is a **laboratory validation case**: it is what you would apply on a test rig to compare against strain gauges (Milestone 6).

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10. Boundary conditions and finite-element model

- **BC:** clamped root, u = v = w = ψ_x = ψ_y = 0 at x = 0.
- **Mesh:** 40 × 10 four-node FSDT elements (25 × 25 mm), 451 nodes, **2255 DOF** (2200 free). Milestone 2 shows this mesh is within 0.08 % (deflection) and 0.32 % (first six frequencies) of the converged solution.



Mesh

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11. Summary of decisions and changes from the planning defaults

Item	Planning default	Final	Why
Laminate	8-ply $[0/45/-45/90]_s$ solid, 1 mm	**Sandwich $[0/45/-45/90/C]_s$, 22.75 mm**	A 1 mm solid plate of 1 m span would deflect ~2 m under 1 g, which is physically impossible. The sandwich matches the real airfoil's skin bending stiffness.

Item	Planning default	Final	Why
G_{23} (T300)	3.0 GPa	3.46 GPa	Derived from transverse isotropy, ν_{23} = 0.49
Shear correction	k = 5/6	Energy-equivalent (0.095 / 0.092)	k = 5/6 is 9× wrong for sandwiches
Limit load factor	3.8	**5.11**	Gust-critical aircraft
Material, geometry, loads	as planned	as planned	—

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12. How to verify in MATLAB

```
>> cd uav_shm_dt
>> setup_paths
>> run_milestone1          % ~5 s
```

You should see these numbers:

Check	Expected
d_eq	22.26 mm
I_o	2.731 kg/m²
A_{11}, D_{11}	78.1608 MN/m, 9703.16 N·m
k_x, k_y	0.0954, 0.0919
As_{11} (energy k) vs Allen	0.5564 vs 0.5463 MN/m
V_S1, V_A	12.66, 24.67 m/s
Design n_+	5.11 (gust at V_D)
LC4 root moment	52.424 N·m
Semi-wing mass	0.6828 kg

Six figures are written to `results/figures/M1_*.png`.

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13. References

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